POKHARA UNIVERSITY

Level: Bachelor
Programme: BE
Course: Physics

Semester: Fall

Year : 2022
Full Marks: 100
Pass Marks: 45
Time : 3hrs.

Candidates are required to give their answers in their own words as far as practicable.

The figures in the margin indicate full marks.

Attempt all the questions.

1. a) What is a compound pendulum? Derive its time period and explain the conditions for the maximum and minimum time period of this pendulum. 9

   b) What will be ratio of intensities of two waves $y_1 = 5 \sin(0.3t-20x)$ and $y_2 = 4 \sin(0.4t-25x)$, where the value of 'x' and 'y' are in cm and 't' is in second? 6

2. a) What are coherent sources? How are they realized in practice? Give with necessary theory, the Newton's rings Method for the determination of wavelength of monochromatic light when space between lens and glass plate is filled with liquid of refractive index μ. 9

   b) Light of wavelength 5000 A is incident normally on a plane transmission grating. Find the difference in the angle of deviation in the first and third order spectra. The number of lines per cm on grating is 6000. 6

3. a) Define electric field and potential due to electric monopole. Derive the expression for electric field intensity at any point due to electric dipole and specify for a point at axial line and at equatorial line. 9

   b) Obtain the time constant of a capacitor in RC circuit such that the current through the resistor increases to one third of its peak value in 5 seconds. 6

4. a) What is the use of Biot-Savart's law and Ampere's law? State Biot-Savart's law. Use it to find the magnetic field due to an infinitely long straight wire. 9

   b) What are the (i) mean time between collisions? (ii) The mean free path for free electrons in copper? (Given $n=8.4\times10^{28}m^{-3}$, $\rho=1.7\times10^{-8}\Omega m$, $v_{avg}=1.6\times10^{6}$ m/s.) 6

5. a) Write the Maxwell's equation in integral form and convert them into differential form. 9

   b) A 40mH inductor and 100μF capacitor form oscillating circuit. What is the peak value of current if the initial charge is 40μC? 6

6. a) What is wave function? Derive the one-dimensional Schrodinger's time dependent equation. 9

   b) Explain briefly about different types of biasing of PN junction diode. 6

7. Write short notes on: **(Any two)** 2×5

   a) Principle of working of Laser

   b) Hall's effect.

   c) Production of ultrasound.